Lecture 1

1. Basic Definitions

- **Variable**: a characteristic or attribute that can assume different values.
- **Data**: values or measurements that variables can take.
- **Random variable**: a variable whose values are determined by chance.

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2. Population vs Sample

- **Population**: all subjects under study.
- **Sample**: a selected group from the population.

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3. Types of Statistics

1. **Descriptive statistics**:
    - Organize, summarize, present data.
    - Example: average age, income from census.
2. **Inferential statistics**:
    - Generalize from sample → population, perform hypothesis tests, predictions.
    - Uses **probability**.
    - Example: study sample of people with high-calorie diets → infer risk of memory loss.

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4. Variables & Data Types

A. Qualitative (Categorical)

- Values are **categories**.

B. Quantitative (Numerical)

- Values can be **counted or measured**.

Quantitative Subtypes

1. **Discrete –** countable, integer values.
    - Examples: number of children, students, calls.
2. **Continuous –** can take infinite values between two points.
    - Examples: height 170.5, weight 65.2, temperature 36.6°C.

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5. Motivation / Purpose of Statistical Study

- Collect data for variables of interest.
- Organize data into **frequency distributions**.
- Present data using **charts and graphs**.

Workflow: Collect → Organize → Present → Observe → Conclude.

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6. Frequency Distributions

A. Categorical Frequency Distribution

- Used for **qualitative data** (categories).
- Example: blood type distribution (A, B, O, AB).

Steps:

1. Create table with **Class, Tally, Frequency, Percent**
2. Tally each occurrence
3. Count frequency
4. Calculate % = (frequency / total) × 100
5. Check sum of percentages ≈ 100%

B. Grouped Frequency Distribution

- Used for **quantitative data with large range**.
- Steps:
    1. Determine classes:
        - Find **highest (H) & lowest (L)** → range R = H - L
        - Choose **number of classes** (5–20)
        - Class width = R ÷ # of classes → round up
        - Determine **lower & upper class limits**
        - Determine **class boundaries**: lower −0.5, upper +0.5
    2. Tally data
    3. Count **frequencies**
    4. Optional: calculate **cumulative frequencies**

C. Cumulative Frequency Distribution

- Shows **number of data values ≤ a specific value**.
- Calculated by **adding frequencies cumulatively**.

7. Reasons for Frequency Distributions

1. Organize data meaningfully تنظيم البيانات بطريقة مفهومة ومرتبة
2. Show **shape of distribution** عرض شكل التوزيع
3. Facilitate calculations (mean, median, spread) تسهيل الحسابات مثل الوسط الحسابي، الوسيط، والانحراف
4. Enable charts/graphs presentation تمكين عرض البيانات باستخدام المخططات والرسوم البيانية
5. Allow comparisons among datasets تمكين المقارنات بين مجموعات بيانات مختلفة

Data Presentation

Histogram

- Graph made of **adjacent vertical bars**.
- **X-axis:** class boundaries.
- **Y-axis:** frequencies.
- Height of bar = frequency of that class.

Frequency Polygon

- Uses **midpoints** of classes instead of bars.
- Connect points with straight lines.
- Must **start and end on the X-axis** (extend one class before and after).

👉 **Midpoint Formula:**

Midpoint = (Highest boundary + Lowest boundary) / 2

Ogive (Cumulative Frequency Graph)

- Based on **cumulative frequencies**.
- **X-axis:** class boundaries.
- **Y-axis:** cumulative frequencies.
- Plot cumulative points and connect them.
- Shows “less than or equal to” values (running total).
- Starts on X-axis at the lower boundary of the first class.

Steps to Construct (all three)

1. Draw and label X and Y axes.
2. On X-axis:

   - Use **class boundaries** (Histogram, Ogive).
   - Use **midpoints** (Frequency Polygon).
3. On Y-axis:
   - Use frequency (Histogram, Polygon)
   - Use Cumulative frequency (Ogive)
4. Plot frequencies:
   - Bars for Histogram.
   - Lines for Polygon and Ogive.

⚡ Quick Use:

- **Histogram** → shows frequency distribution with bars.
- **Polygon** → easier to compare distributions, uses midpoints.
- **Ogive** → shows cumulative totals (how many values are ≤ some point).

Lecture 2

Statistics vs. Parameters

- A **statistic** is a numerical measure obtained from **sample data**.
- A **parameter** is a numerical measure obtained from **population data**.
- Statistics describe samples, while parameters describe populations.

Sample Mean vs. Population Mean

- The **mean** is the sum of all data values divided by the total number of values.
- The **sample mean** ($\bar{X}$) uses data from a sample and is a **statistic**.
- The **population mean** (μ) uses all population data and is a **parameter**.
- The mean is usually not an actual data value.

Mean for Grouped Data

- For grouped data, the mean is found by multiplying each class midpoint by its frequency, summing these products, and dividing by the total frequency.
- Formula:

$$\bar{X} = \frac{\Sigma(f \times X_m)}{\Sigma f}$$

where X_m is the midpoint and f is the frequency.

The Median

- The **median (MD)** is the middle value when data are arranged in ascending order.
- If the number of data values is **odd**, the median is the middle value.
- If **even**, it is the average of the two middle values.

The Mode

- The **mode** is the value that occurs most frequently in a data set.
- A data set can be:
  - **Unimodal** (one mode)
  - **Bimodal** (two modes)
  - **No mode** (no repeated values)
- For grouped data, the **modal class** is the class with the highest frequency.
- For categorical data, the mode is the category with the greatest frequency.

The Midrange

- The **midrange (MR)** is the sum of the lowest and highest data values divided by 2.
- Formula:

$$MR = \frac{\text{Lowest value} + \text{Highest value}}{2}$$

The Weighted Mean

- The **weighted mean** is found by multiplying each value by its weight, summing these products, and dividing by the sum of the weights.
- Formula:

$$\bar{X}_w = \frac{\Sigma(w \times X)}{\Sigma w}$$

where w= weight and X= value.

Summary of Measures of Central Tendency

Measure	Definition	Symbol
Mean	Sum of all values divided by the total number of values	$\bar{X}$ or μ

Measure	Definition	Symbol
Median	Middle point in an ordered data set	MD
Mode	Most frequent data value	None (symbol varies)
Midrange	(Lowest + Highest) ÷ 2	MR

🔷 2. The Range (المدى)

The **range (R)** measures the **difference between the highest and lowest** value:

$$R = \text{Highest value} - \text{Lowest value}$$

🔷 3. Population Variance and Standard Deviation

They measure how far each data value is from the **mean (المتوسط).**

Symbols:

- σ^2: population variance
- σ: population standard deviation
- μ: population mean
- N: population size

Formulas:

$$\sigma^2 = \frac{\sum(X-\mu)^2}{N}$$
$$\sigma = \sqrt{\frac{\sum(X-\mu)^2}{N}}$$

🔷 4. Steps to Find Population Variance & Standard Deviation

Step	Description
1	Find the mean: $\mu = \frac{\sum X}{N}$
2	Find deviations: $X - \mu$
3	Square each deviation: $(X-\mu)^2$
4	Find sum of squares: $\sum(X-\mu)^2$
5	Divide by N: gives **variance**
6	Take square root: gives **standard deviation**

🔷 7. Sample Variance and Standard Deviation

Used when we have **a sample (عينة)** instead of the whole population.

Formulas:

$$s^2 = \frac{\sum(X - \bar{X})^2}{n - 1}$$

$$s = \sqrt{\frac{\sum(X - \bar{X})^2}{n - 1}}$$

🔷 9. Shortcut Formulas

For faster calculation:

$$s^2 = \frac{n(\sum X^2) - (\sum X)^2}{n(n - 1)}$$

$$s = \sqrt{\frac{n(\sum X^2) - (\sum X)^2}{n(n - 1)}}$$

⚠️ Note:

- $\sum X^2$: square each value first, then sum.
- $(\sum X)^2$: sum first, then square.

✅ Example result: variance = 13.1, standard deviation = 3.6

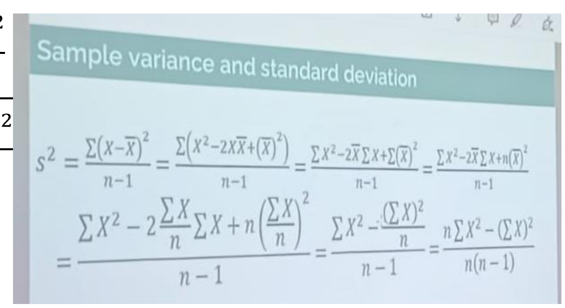
🔷 10. Grouped Data

Used for frequency tables (البيانات المجمعة).

Formulas:

$$s^2 = \frac{n(\sum f X_m^2) - (\sum f X_m)^2}{n(n - 1)}$$

$$s = \sqrt{\frac{n(\sum f X_m^2) - (\sum f X_m)^2}{n(n - 1)}}$$



Where:

- X_m = midpoint of each class
- f = frequency

🔷 11. Steps for Grouped Data

Step	Description
1	Find class midpoints
2	Multiply f × X_m
3	Multiply f × X_m^2
4	Find Σf, ΣfX_m, ΣfX_m^2
5	Substitute into formula
6	Take √ to find s

🔷 13. Summary of Measures of Variation

Measure	Definition	Symbol
Range	Highest − Lowest	R
Variance	Average of squared distances from mean	σ^2, s^2
Standard Deviation	√Variance	σ, s

🔷 14. Coefficient of Variation (CVar)

Shows variation **relative to the mean** (كنسبة مئوية).

$$CVar = \frac{s}{\bar{X}} \times 100 \text{(sample)}$$

$$CVar = \frac{\sigma}{\mu} \times 100 \text{(population)}$$

Example:

- Sales: $s = 5, \bar{X} = 87$→ CVar = 5.7%
- Commissions: $s = 773, \bar{X} = 5225$→ CVar = 14.8%

✅ Commissions are more .

🔷 15. Chebyshev's Theorem (نظرية تشيبيشيف)

Applies to **any** data shape.
It tells what **proportion** of values lie within *k* standard deviations from the mean.

$$P = 1 - \frac{1}{k^2}, k > 1$$

Examples:

- For $k = 2$: at least **75%** of data

- For $k = 3$: at least **88.9%** of data

✅ Works for any distribution (not only normal).

Example 1:
House prices mean = $50,000, s = $10,000
→ 75% between $30,000 and $70,000

🔷 16. Empirical (Normal) Rule

Applies only when data are **normal (bell-shaped)**.

Range	% of data
±1σ	≈ 68%
±2σ	≈ 95%
±3σ	≈ 99.7%

Example:
Mean = 480, σ = 90

- 68% between 390 and 570
- 95% between 300 and 660
- 99.7% between 210 and 750

✅ Empirical rule is more accurate than Chebyshev's (only for normal distributions).

Lecture 3

1. Probability Experiment (تجربة احتمالية)

A **probability experiment** is any process that produces results (outcomes) by chance.

- **Outcome (ناتج):** Result of one trial (like flipping a coin once).
- **Sample Space (فضاء العينة):** The set of all possible outcomes.

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2. Event (حدث)

An **event** is one or more outcomes from the experiment.

- **Simple Event (حدث بسيط):** One outcome (e.g., rolling a 6).
- **Compound Event (حدث مركّب):** Two or more outcomes (e.g., getting an odd number when rolling a die → 1, 3, 5).
- **Equally Likely Events (أحداث متساوية الاحتمال):** Each has the same chance of happening.

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3. Sample Space Examples

- **Two Dice:**
  Each die has 6 sides, so total outcomes = 6 × 6 = 36.
  Example: (1,1), (1,2), ..., (6,6).
- **Drawing One Card:**
  A deck has **4 suits** × **13 cards each** = **52 outcomes**.

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4. Tree Diagram (مخطط الشجرة)

A **tree diagram** is used to visualize all possible outcomes.
Example: The gender of three children → Each child can be Boy (B) or Girl (G).
The possible outcomes are:
BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG (8 outcomes).

🎲 Classical Probability (الاحتمال الكلاسيكي)

It assumes all outcomes in the sample space are **equally likely**.

Formula:

$$P(E) = \frac{n(E)}{n(S)}$$

where:

- $n(E)$= number of favorable outcomes (عدد النتائج المطلوبة)
- $n(S)$= total number of possible outcomes (إجمالي النتائج الممكنة)

🧩 1. Addition Rules (قواعد الجمع)

🔷 a. Mutually Exclusive Events (الأحداث المتنافية)

These events **cannot happen at the same time**.
Example: Getting a head and a tail in one coin toss.

Rule:

$$P(A \text{ or } B) = P(A) + P(B)$$

Example:
A die is rolled, find the probability of getting 2 or 5:

$$P(2 \text{ or } 5) = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

🔷 b. Not Mutually Exclusive (أحداث غير متنافية)

These events **can occur together**.
Rule:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Example:
Drawing a card that is a king or a heart:

$$P(K \text{ or } \heartsuit) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52}$$

🔷 **c. For Three Events (لثلاثة أحداث)**

General rule (Inclusion–Exclusion Law):

$$\begin{aligned} P(A \text{ or } B \text{ or } C) &= P(A) + P(B) + P(C) \\ &-P(A \text{ and } B) - P(A \text{ and } C) - P(B \text{ and } C) \\ &+P(A \text{ and } B \text{ and } C) \end{aligned}$$

📘 *Arabic note:* نطرح التقاطعات المزدوجة ثم نضيف الثلاثي في النهاية.

🎲 **2. Multiplication Rules (قواعد الضرب)**

🔷 **a. Independent Events (أحداث مستقلة)**

Events are **independent** if one does not affect the other.
Rule 1:

$$P(A \text{ and } B) = P(A) \times P(B)$$

Example:
A coin is flipped and a die is rolled.

$$P(H \text{ and } 4) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12} \approx 0.083$$

🔷 **b. For Three Independent Events (لثلاثة أحداث مستقلة)**

Extended Rule:

$$P(A \text{ and } B \text{ and } C) = P(A) \times P(B) \times P(C)$$

Example:
If 46% of people have stress weekly, probability that 3 people do:

$$(0.46)^3 = 0.097$$

➡️ 9.7% chance.

🔷 c. Dependent Events (أحداث معتمدة)

Events are **dependent** if the outcome of the first affects the second.
Examples:

- Drawing a card and not replacing it.
- Selecting a ball without replacement.

🔷 d. Conditional Probability (الاحتمال الشرطي)

The probability that **B occurs given that A has already occurred.**

Notation: $P(B \mid A)$

Formula:

$$P(B \mid A) = \frac{P(A \text{ and } B)}{P(A)}$$

📘 *Arabic note:* نستخدمها لما الحدث التاني يعتمد على الأول.

🔷 e. Multiplication Rule 2 (لأحداث معتمدة)

When events are **dependent**:

$$P(A \text{ and } B) = P(A) \times P(B \mid A)$$